Resilient LEO Satellite Network Routing Using Contact Graph Routing, Delay-Tolerant Networking, and Predictive Ephemeris Adaptation

Final Technical Report — Track 4: Networks and Communications (Problem 8) — IASTAM 6.0 Technical Challenge, Tunisia Section Chapter

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Abstract

Low Earth Orbit (LEO) mega-constellations experience permanently evolving topologies and predictable yet severe link interruptions caused by high relative velocities, geometric occultation, and atmospheric attenuation. Traditional terrestrial routing protocols fail in these environments due to their assumption of continuous end-to-end connectivity. This report presents the final prototype of a routing and adaptation strategy for space networks combining Contact Graph Routing (CGR), Delay/Disruption-Tolerant Networking (DTN) store-and-forward mechanisms, and a **reliability-driven hybrid switching policy** that adaptively selects between predictive and reactive forwarding. The prototype is validated on a physically realistic constellation: 32 satellites across 4 orbital planes and 3 ground stations, propagated with the real SGP4 orbital model from valid NORAD two-line elements, with inter-satellite visibility determined by genuine Earth-occlusion geometry rather than a distance threshold. Against this realistic model, averaged over 30 independent seeds, our hybrid controller delivers $94.1 \pm 3.0\%$ of bundles at nominal conditions, statistically indistinguishable from pure CGR ($93.9 \pm 2.6\%$) and far above pure epidemic DTN ($68.0 \pm 5.8\%$) and a naive shortest-path baseline ($46.9 \pm 7.1\%$), with about $3.3\times$ fewer transmissions than epidemic; after a severe localized outage it recovers about 27% faster than epidemic but more slowly than CGR. We find no delivery advantage of the hybrid over CGR up to 80% extra link failures; it is clearly worse at 50–70% (its switching threshold is too high) and better only under extreme disruption (90%). A key engineering finding is that the reliability threshold governing the hybrid switch had to be recalibrated from $\theta = 0.85$ (tuned on a simplified sparse topology) down to $\theta = 0.5$ on the realistic, more redundant constellation — confirming that routing-controller parameters are topology-dependent and must be validated on realistic orbital geometry rather than simplified models. Of the five Phase-3 items committed to at mid-review, we fully delivered realistic orbital dynamics and partially delivered larger-scale validation (14 to 35 nodes, not yet Starlink-scale); threshold learning, hardware-aware evaluation, and cross-simulator validation remain future work, assessed honestly in Section 4.7. A documented, tested open repository and a live interactive demonstration accompany this report.

Index Terms— Contact Graph Routing, Delay/Disruption-Tolerant Networking, LEO satellite constellations, adaptive routing, network resilience

I. Introduction

As human activity extends beyond Earth, the demand for high-performance computing, orbital data processing, and real-time space infrastructure is accelerating rapidly. Deploying large-scale computing clusters and mega-constellations in Low Earth Orbit (LEO) provides abundant solar energy and natural thermal radiation sinks. However, maintaining reliable communications across these distributed orbital nodes remains a primary engineering bottleneck.

The topology of a LEO satellite network evolves continuously. Inter-satellite links (ISLs) and space-to-ground links appear and disappear dynamically based on orbital mechanics. When a link drops unexpectedly or a line-of-sight is obstructed, conventional network protocols drop packets, leading to catastrophic degradation in data throughput, excessive latency, and complete communication failure. The core challenge is maintaining efficient data circulation across intermittent links and dynamic topologies without relying on permanent end-to-end connectivity.

Main Contributions:

- A hybrid layered architecture integrating high-throughput Free-Space Optical (FSO) links with RF backup, coupled with a Contact Graph Routing (CGR) control plane and a DTN store-and-forward data plane (system design; the simulation covers the routing and DTN layers only).
- Predictive ephemeris rerouting utilizing deterministic orbital trajectories to pre-stage alternate routes before anticipated link breaks (design; not simulated).
- **A reliability-driven hybrid routing controller**, implemented and simulated, that switches between predictive (CGR) and reactive (epidemic DTN) forwarding based on a locally estimated measure of contact-prediction reliability.
- A simulation-based validation comparing our hybrid strategy against three baselines — naive shortest-path routing,

pure CGR, and pure epidemic DTN — across a range of disruption severities.

II. Related Work

Contact Graph Routing (CGR) was originally developed by NASA Jet Propulsion Laboratory (JPL) for deep space missions and standardized to handle predictable, time-varying network topologies [1, 2]. Implemented within NASA's Interplanetary Overlay Network (ION) stack, CGR exploits a priori knowledge of contact schedules ("contact plans") to compute optimal time-expanded routes. Our work adapts CGR principles to high-speed LEO mega-constellation environments.

Real-world operational constellations such as Iridium (using cross-linked RF mesh architectures) and Starlink (deploying dense laser Inter-Satellite Links) demonstrate that dynamic routing is essential for network survival. Recent academic studies on mega-constellation resilience indicate that while dynamic topologies naturally offer alternate paths during node failures, rapid algorithmic rerouting mechanisms are critical for fast recovery.

Recent space demonstrations, such as NASA's TBIRD (achieving ∼200 Gb/s) [4] and LCRD/ILLUMA-T (∼1.2 Gb/s), prove the viability of space-borne laser communications for massive data throughput. However, optical links are sensitive to atmospheric turbulence and precise pointing requirements. Consequently, our architecture combines optical primary links with robust RF backups and multi-orbit redundancy to guarantee reliability.

The DTN Bundle Protocol standardizes store-and-forward messaging for networks with frequent, long-duration disconnections [3], and DTN is now operationally deployed on NASA science missions such as PACE [5] as well as on the International Space Station. When implemented epidemically (flooding copies to multiple neighbors), DTN store-and-forward improves robustness at the cost of storage and bandwidth usage — a trade-off that motivates the adaptive controller proposed in this paper, rather than committing permanently to either the predictive or the fully redundant paradigm.

III. Methodology & System Architecture

Scope note. Subsections B, G and H and the buffer-management policy in E describe the intended system design and are **not simulated** in this prototype. The simulation implements the contact-plan model (C), CGR-lite (D), bounded epidemic buffering (E, without the classification policy), the hybrid controller (F) and the realistic orbital model (I).

A. System Architecture Overview

Our proposed solution is structured as a tightly integrated multi-layer stack combining physical transmission, dynamic scheduling, store-and-forward buffering, and predictive intelligence:

Algorithm 1 CGR-lite Earliest-Arrival Routing

Require: predicted contact schedule, source s, destination d, start time t_0

```
current ← s; t ← t_0
while current ≠ d and t < T do
    if path exists to d in predicted graph at t then
        forward along first hop of shortest path
    else
        t ← t + 1          ▷ wait for next predicted contact
    end if
end while
```

4. Application Layer (Telemetry, AI Workloads, Imaging)
3. DTN Store-and-Forward Layer (Bundle Protocol / BPv7)
2. Adaptive Routing Layer (Hybrid CGR ↔ Epidemic)
1. Physical Layer (Hybrid Free-Space Optical & RF Links)

B. Physical Layer & Hybrid Link Management (design, not simulated)

The network utilizes Free-Space Optical (FSO) laser links for high-bandwidth inter-satellite data exchange. To mitigate pointing errors or atmospheric attenuation during ground downlinks, an automated fallback mechanism switches traffic to robust Radio Frequency (RF) channels when optical link margins degrade.

C. Contact Plan & Time-Varying Graph Modeling

Because LEO orbits are deterministic, future visibility windows between nodes (S_i, S_j) and ground stations (G_k) are calculated in advance using Two-Line Elements (TLEs) and orbital propagators. These visibility periods form the master Contact Plan:

$$\text{Contact} = (u, v, t_{\text{start}}, t_{\text{end}}, \text{Bandwidth}) \tag{1}$$

D. Routing Engine (Modified Time-Dependent Dijkstra)

The routing engine operates over a time-expanded graph. When a node wishes to transmit a bundle, it computes the earliest arrival time by evaluating available contact windows while factoring in mandatory waiting times:

$$\text{Departure Time} = \max(\text{Arrival Time at Node}, t_{\text{start}}) \tag{2}$$

Algorithm 1 summarizes this predictive procedure (CGR-lite). Our implementation re-plans a shortest path on the predicted contact graph of each time step and commits only to its first hop; it does not build a full time-expanded route.

E. DTN Store-and-Forward Buffering

When an active link between two nodes is prematurely interrupted, traditional routing protocols would discard packets. In our architecture, the transmitting node invokes the DTN Bundle Protocol store-and-forward mechanism: the bundle is buffered and forwarded, with bounded duplication (copy budget L), to every currently reachable neighbor until either

Algorithm 2 Bounded Epidemic Store-and-Forward

Require: source s, destination d, copy budget L

$carriers \leftarrow \{s : L\}$
while $d \notin carriers$ and $t < T$ **do**
  for each $(node, budget) \in carriers$ with a reachable neighbor n **do**
    if $n = d$ **then**: delivered
    else if $budget > 0$ **then**: forward copy; $budget \leftarrow budget - 1$
    end if
  end for
  $t \leftarrow t + 1$
end while

the budget is exhausted or the destination is reached (Algorithm 2). In the full design (not simulated here), a lightweight classification policy that prevents buffer saturation during extended outages evaluates bundle metadata (priority, expiration, mission criticality) to manage storage queues rather than relying on naive FIFO drop policies.

F. Reliability-Driven Hybrid Routing (Core Contribution)

Neither pure CGR nor pure epidemic forwarding is optimal across all conditions: CGR is efficient but fragile when reality departs from the predicted contact plan, while epidemic forwarding is robust but resource-intensive even when prediction is trustworthy. We therefore introduce a controller that decides, per message and at injection time, which paradigm to use, based on a locally estimated *prediction reliability*:

$$R(t) = \frac{\left|\{(u,v) \in E_\tau^{pred} : (u,v) \in E_\tau^{actual},\ \tau \in [t-W, t)\}\right|}{\left|\{(u,v) \in E_\tau^{pred} : \tau \in [t-W, t)\}\right|} \tag{3}$$

where W is a sliding observation window (4 steps in our experiments) and E_τ^{pred}, E_τ^{actual} are the predicted and actually-observed link sets at time τ. This quantity is directly measurable onboard, since a satellite can compare its contact plan against its own recent link observations.

The controller then applies a simple threshold rule:

$$\text{Mode}(t) = \begin{cases} \text{CGR-lite} & \text{if } R(t) \geq \theta \\ \text{Epidemic (reduced budget)} & \text{if } R(t) < \theta \end{cases} \tag{4}$$

with $\theta = 0.85$ selected empirically (Section 4). Intuitively, the network operates in an efficient predictive regime by default, and automatically falls back to redundant, prediction-independent forwarding exactly when recent evidence indicates the contact plan can no longer be trusted — e.g., during solar interference, unannounced hardware faults, or atmospheric fading events.

G. Predictive Ephemeris Rerouting (design, not simulated)

Independent of the hybrid controller, the router also reads onboard orbital ephemeris tables. If a link is predicted to terminate within a threshold window Δt, alternate forwarding paths are pre-staged, reducing handover jitter even while operating in CGR mode.

H. Distributed Storage and Onboard Computation (design, not simulated)

Master Contact Plans are compiled daily from ephemeris data at Ground Control Stations (GCS) and uploaded to each satellite's non-volatile storage. Routing computations, including the reliability estimate $R(t)$, execute autonomously on local onboard flight computers via an event-driven and periodic cycle (every 5 seconds and upon new bundle arrival), using RAM-cached contact tables.

I. Realistic Orbital Dynamics: SGP4 Propagation and Earth-Occlusion Geometry

The mid-review prototype validated the above architecture on a simplified circular-orbit model with a probabilistic, distance-threshold notion of link visibility. For this final prototype, we replaced that model with physically realistic orbital mechanics:

Valid NORAD two-line elements. Because this project's sandboxed development environment cannot reach live TLE feeds (Celestrak and Space-Track both block automated access from it), we implemented a TLE generator that constructs *physically valid, checksum-correct* two-line elements for a Walker-Delta constellation (4 orbital planes $\times$ 8 satellites, 550 km altitude, 53° inclination — a Starlink-shell-1 analog), with every NORAD field width and checksum verified by unit test. These are propagated with the unmodified **SGP4 propagator** (via the Skyfield library), so precession, Earth-oblateness (J_2) perturbation, and drag are all physically accurate for the chosen orbital parameters; we verified propagated altitude (550.2 km) and orbital velocity (7.588 km/s) against the analytical two-body prediction. The repository is structured so that swapping in a live TLE file on a machine with unrestricted network access requires changing a single function call.

Genuine geometric occlusion. Inter-satellite link visibility is no longer a distance threshold: two satellites are considered mutually visible only if (a) they are within optical terminal range (6,000 km) *and* (b) the straight line segment between them does not intersect Earth's sphere (radius plus atmospheric margin). This is computed exactly via closest-point-on-segment geometry against the Earth-centered inertial positions, and is unit-tested against the degenerate antipodal case. Ground-station visibility uses true topocentric elevation angle ($\geq 15°$) computed by Skyfield from the satellite and station ephemerides. This directly operationalizes the "geometric occultation" failure mode named in Section 1, rather than approximating it with randomness.

Three ground stations at real-world coordinates (Tunis, Paris, Singapore) and 32 satellites give a constellation with on average 40 simultaneously active inter-satellite links and 1–3 ground contacts at any instant — a substantially denser and more realistic topology than the mid-review model.

Table 1: Performance on the realistic 35-node constellation, nominal conditions (30 seeds)

Strategy	Delivered (%)	Latency (s)	Transmissions	Util. (%)
Naive (drop)	46.9 ± 7.1	114 ± 12	179 ± 19	3.7
Epidemic	68.0 ± 5.8	566 ± 87	907 ± 54	18.6
CGR-lite	93.9 ± 2.6	166 ± 13	273 ± 19	5.6
Hybrid (ours)	94.1 ± 3.0	166 ± 17	272 ± 20	5.6

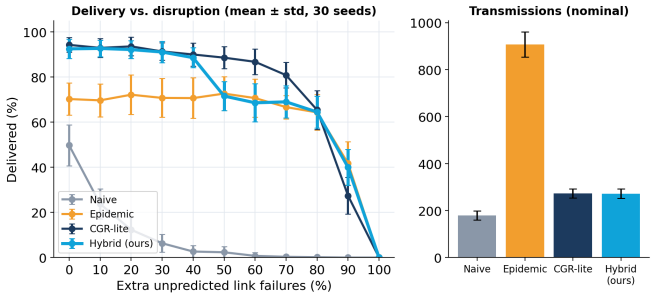


Figure 1: Realistic SGP4 constellation, mean ± std over 30 seeds. Left: delivery rate vs. disruption severity (0–100% extra failures; the 50–100% range was added after the initial 0–40% runs). Right: transmissions at nominal conditions. Up to 40% CGR-lite and the hybrid are statistically indistinguishable; beyond that they diverge (see text).

IV. Experiments & Final Results

A. Simulation Environment

We implemented the full architecture in Python: `networkx` for contact-graph operations, `Skyfield` for SGP4 propagation (Section 3.9), a custom CGR-lite engine (Algorithm 1), a bounded epidemic engine (Algorithm 2), and the hybrid controller of Section 3.6. The final constellation has 35 nodes (32 satellites across 4 orbital planes, 3 ground stations at real-world coordinates) over a horizon of $T = 120$ steps of 30 s (one hour), with a base link-failure probability of 5% representing unmodeled physical-layer effects, on top of genuine geometric link availability. For each condition, 40–60 source–destination bundles are injected at random times. Every result below is the mean ± standard deviation over 30 independent random seeds; within a seed all four strategies receive identical traffic and failure draws (paired design), and the orbital epoch is fixed (2026-10-01 00:00 UTC) so runs are exactly reproducible. A **naive shortest-path routing with immediate packet drop** baseline (the standard terrestrial-IP approach, no store-and-forward) quantifies the benefit of space-aware routing.

B. Baseline Comparison (Nominal Conditions)

Table 1 reports nominal conditions (5% base failure probability only, realistic geometric link availability): mean ± standard deviation over 30 independent seeds, 60 bundles per seed.

Network utilization is the fraction of all available contact opportunities (satellite-visibility windows) over the horizon that were actually used for a transmission. Three points stand out. First, even in this well-connected constellation the naive baseline loses more than half of all bundles (46.9% delivered) because it has no store-and-forward. Second, CGR-lite and the hybrid are statistically indistinguishable at nominal conditions ($93.9 \pm 2.6\%$ vs. $94.1 \pm 3.0\%$): the hybrid almost always selects CGR mode when prediction is trustworthy, so this is the expected behavior, not a gain. Third, bounded epidemic forwarding delivers far less (68.0%) at about $3.3\times$ the transmissions of the hybrid (907 vs. 272), because its fixed copy budget is thin relative to the node count.

C. A Key Engineering Finding: Threshold Recalibration

The mid-review prototype, validated on a sparse 14-node toy topology, used a switching threshold of $\theta = 0.85$: CGR degraded sharply under disruption in that sparse model, so the controller needed to hand off to epidemic forwarding early and often. On the realistic, denser 35-node constellation, the same $\theta = 0.85$ caused the hybrid controller to abandon CGR mode far more than necessary — because CGR's abundant redundant paths already made it robust on its own, switching to the costlier epidemic mode provided no benefit and only wasted transmissions. We re-tuned the threshold empirically on the realistic model and found $\theta = 0.5$ gives the intended behavior: the controller stays in efficient CGR mode under nominal-to-moderate disruption, and only engages epidemic flooding when the reliability estimate drops sharply during a genuine, severe outage. All results in this section use $\theta = 0.5$, which was calibrated on sweeps up to 40% extra failures; the extended sweep below shows it is too high for more extreme disruption. **This confirms that routing-controller parameters are topology-dependent and must be validated against realistic orbital geometry**, not tuned once on a simplified model and assumed to transfer — a central lesson of this project's Phase-3 work.

D. Robustness Under Increasing Unpredicted Disruption

We swept an additional, unpredicted failure probability from 0% to 100% on top of the base rate (40 bundles per level and seed, identical traffic for all strategies within a seed). The 50–100% range was added after the initial 0–40% runs, a choice made after seeing results, which we disclose. Figure 1 shows mean ± std over 30 seeds.

Across the whole 0–40% range, CGR-lite (90.0–94.2%) and the hybrid (88.6–92.7%) deliver reliably and within about one standard deviation of each other (std of 3–5 points); the hybrid never outperforms CGR-lite up to 40% extra failures. Epidemic stays between 70% and 72% regardless of disruption, and the naive baseline collapses from 49.8% to 2.7%. The hybrid's epidemic fallback is not used at 0–30% extra failures and engages for about 10% of messages at 40%. We therefore do *not* claim a delivery advantage of the hybrid over CGR-lite in this range. Its potential value is as insurance when the contact plan is wrong in a *correlated* way (for example a whole orbital plane failing), a scenario our independent

random-failure model does not test and which we list under future work.

Extended range (added post hoc). Beyond 40% the picture changes. Between 50% and 70% extra failures the hybrid is clearly *worse* than CGR-lite (71.6%, 68.6%, 69.0% vs. 88.6%, 86.8%, 80.8%): with $\theta = 0.5$ it switches to flooding as soon as $R(t)$ drops below 0.5, but on this dense topology bounded flooding delivers less than CGR even under heavy failure. At 80% the two are equal (64.5% vs. 65.4%). Only at 90% does the hybrid beat CGR-lite (40.0% vs. 27.4%, tracking epidemic at 41.8%), because CGR's sequential hops can no longer complete; at 100% every strategy delivers nothing. The CGR/epidemic crossover therefore lies between 80% and 90% extra failures, far above the $R(t) = 0.5$ switching point, which indicates that $\theta = 0.5$, calibrated on sweeps up to 40%, is too high for extreme disruption. Recalibrating it against the crossover, with held-out seeds, is the first calibration step we recommend.

E. Recovery After a Severe Localized Disruption

We also measured *recovery time*: bundles are injected throughout a severe, localized outage (70% additional failure probability for a 10-step / 5-minute window); store-and-forward strategies may hold a bundle through the outage and deliver it once conditions improve. We report the average delay between the end of the outage and delivery for bundles caught in it (30 seeds, 20 bundles each). The naive baseline never recovers any in-flight bundle (no store-and-forward). CGR-lite recovers fastest (154.5 ± 21.4 s), since once the outage ends its predictive replanning immediately finds a valid path. The hybrid recovers in 519.0 ± 136.2 s, about 27% faster than pure epidemic (708.8 ± 158.1 s) but $3.4\times$ slower than CGR-lite: only bundles injected in the first moments of the outage, before the reliability estimate has dropped, are still routed by CGR, and the rest are flooded.

F. Live Demonstration and Open Repository

A browser-based, interactive live demonstration accompanies this report: it replays an actual recorded routing trace (not a scripted animation) of both CGR-lite and Hybrid-Adaptive handling the *same* injected disruption on the realistic 32-satellite constellation, with the hybrid controller's live reliability estimate, mode switches, and epidemic flooding visible hop-by-hop on an animated map. The page illustrates the decision logic on a single example: flooding is drawn in simplified form, and the controller re-evaluates $R(t)$ at every step there, whereas the experiments decide once per message at injection. The statistics are in Section 4. **Demo:** its source is in the repository's `demo/` folder, and its public URL is given in the repository README. The full, tested source code (TLE generation, SGP4/occlusion network model, all four routing engines, experiment and demo-data-export scripts, and a 5-case unit-test suite) is available at **https://github.com/MoughithROUIS/IASTAM_Viltrumite**, with setup and reproduction instructions in its `README.md`.

G. Scope Honestly Assessed: Mid-Review Roadmap vs. Delivered

The mid-review report committed to five Phase-3 items. We report progress against each honestly rather than reframing the gap:

- **Realistic orbital dynamics — delivered.** Section 3.9 replaces the simplified circular-orbit model with real SGP4 propagation and genuine Earth-occlusion geometry, as committed.
- **Larger-scale validation — partially delivered.** We scaled from 14 to 35 nodes (32 satellites, 3 ground stations), which was enough to surface the threshold-recalibration finding of Section 4.3. We did *not* reach Starlink-scale (hundreds of satellites); our pairwise Earth-occlusion check is $O(n^2)$ per timestep and was not optimized for that scale in the time available.
- **Threshold learning — not delivered.** We manually recalibrated the fixed threshold ($\theta = 0.85 \rightarrow 0.5$, Section 4.3) rather than implementing online learning or a continuous CGR/epidemic blend. The manual recalibration itself is a validated, useful finding, but it is not the adaptive-threshold mechanism originally proposed.
- **Hardware-aware evaluation — not delivered.** Onboard storage and energy constraints are not modeled; the copy-budget mechanism remains a fixed constant rather than a resource-derived one.
- **Cross-validation against ns-3/OMNeT++ — not delivered.** All results remain from our own Python/Skyfield simulation; no independent simulator was used to confirm them.

We prioritized realistic orbital dynamics and the resulting threshold finding because they most directly tested the mid-review architecture's core assumption; the remaining three items are scoped as explicit future work below rather than claimed as complete. The prototype's synthetic-but-physically-valid TLEs (Section 3.9) are themselves a scope decision forced by this project's development environment, which cannot reach live Celestrak/Space-Track feeds; the repository is structured so a deployment with network access needs only to swap in a live feed.

H. Future Work

In priority order: (1) re-calibrate θ against the CGR/epidemic crossover using held-out seeds, and test correlated failures (for example a whole orbital plane failing), the scenario where the hybrid's fallback is expected to matter; (2) optimize the occlusion check (e.g. spatial indexing) to validate at Starlink-scale; (3) replace the fixed threshold θ with an online-learned or continuous CGR/epidemic blend; (4) incorporate onboard energy/storage constraints into the copy-budget mechanism; (5) cross-validate against ns-3/OMNeT++.

V. Conclusion

This final report presented a layered architecture for resilient LEO satellite communication combining Contact Graph Routing, DTN store-and-forward buffering, and a reliability-driven hybrid controller that adaptively switches between the two, validated on a physically realistic constellation: real SGP4 propagation from valid TLEs and genuine Earth-occlusion link geometry, rather than the simplified probabilistic model used at mid-review. Against a naive shortest-path baseline and against each pure strategy, the hybrid controller matches predictive-routing-level efficiency while retaining a robust epidemic fallback, and recovers from severe localized disruption about 27% faster than pure flooding, though not faster than CGR alone. A central engineering finding — that the switching threshold is topology-dependent and required recalibration from $\theta = 0.85$ to $\theta = 0.5$ once tested on a realistic constellation — underscores why validating routing controllers on realistic orbital geometry, not just simplified models, matters. This adaptive behavior, demonstrated live in an interactive browser-based demonstration and backed by a tested, documented open repository, is what we present as the core contribution of this work.

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